SMSystem Documentation

Sales & Inventory Management System

Version 0.3.0

March 2026

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Version: 0.3.0
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Introduction

SMSystem is a comprehensive **Sales & Inventory Management System** designed for multi-branch retail businesses. It provides complete control over inventory, sales, customer relationships, and branch operations.

Key Features

- **Multi-Branch Support**: Manage multiple branches with isolated inventory
- **Point of Sale**: Fast, intuitive checkout with receipt printing
- **Inventory Tracking**: Real-time stock levels with batch tracking
- **Transfer Management**: Move inventory between branches
- **Customer CRM**: Track customer information and loyalty
- **Purchase Orders**: Manage supplier orders and incoming stock
- **Expense Tracking**: Monitor and categorize business expenses
- **Daily Reports**: Generate sales and inventory reports
- **Role-Based Access**: Granular permissions for different user types

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Getting Started

System Requirements

- **Backend**: Go 1.21+, MySQL 8.0+
- **Frontend**: Node.js 18+, npm/yarn
- **Browser**: Chrome 90+, Firefox 88+, Safari 14+

Installation

1. Clone the Repository

```
git clone https://github.com/Jayem09/SMSystem.git
cd SMSystem
```

2. Backend Setup

```
cd backend

# Copy environment file
cp .env.example .env

# Edit .env with your database credentials
# DB_HOST=your-mysql-host
# DB_PORT=3306
# DB_USER=root
# DB_PASSWORD=your-password
# DB_NAME=smsystem

# Install dependencies
go mod download

# Build the server
go build -o smsystem-backend ./cmd/server

# Run the server
./smsystem-backend
```

The server will start on port **8080** by default.

3. Frontend Setup

```
cd frontend

# Install dependencies
npm install

# Start development server
npm run dev

# Build for production
npm run build
```

The frontend will be available at **http://localhost:5173**

First-Time Setup

1. Navigate to the login page
2. Click "Register" to create your first account
3. Fill in your details (name, email, password)
4. Your account will be assigned to a branch
5. Start using the system!

User Roles & Permissions

SMSystem uses role-based access control (RBAC) with the following roles:

Role	Description
super_admin	Full system access, can manage all branches
admin	Branch-level full access, can manage staff and settings
purchasing	Can manage products, inventory, purchase orders
purchaser	Limited purchasing access
cashier	POS access, can process sales
user	Basic access, can view transfers

Role Access Matrix

Feature	super_admin	admin	purchasing	cashier	user
Dashboard	✅	✅	✅	✅	❌
POS	✅	✅	✅	✅	❌
Products	✅	✅	✅	❌	❌
Inventory	✅	✅	✅	❌	❌
Orders	✅	✅	✅	✅	❌
Transfers	✅	✅	✅	✅	✅
Customers	✅	✅	✅	❌	❌
Expenses	✅	✅	✅	❌	❌
Suppliers	✅	✅	✅	❌	❌
Staff	✅	✅	❌	❌	❌
Branches	✅	✅	❌	❌	❌
Reports	✅	✅	❌	❌	❌
Settings	✅	✅	❌	❌	❌

Dashboard

The Dashboard provides an overview of your branch's performance.

Access

- **URL**: `/dashboard`
- **Roles**: All authenticated users except `user`

Features

Summary Cards

- **Today's Sales**: Total revenue for today
- **Orders Today**: Number of orders processed
- **Low Stock**: Products below reorder level
- **Pending Transfers**: Transfers awaiting action

Quick Actions

- **New Sale**: Opens POS for immediate checkout
- **Receive Stock**: Quick access to stock receiving
- **View Reports**: Access daily reports

Recent Activity

- Shows latest orders and activities
- Click to view full details

Point of Sale (POS)

The POS is the primary interface for processing customer transactions.

Access

- **URL**: `/pos`
- **Roles**: super_admin, admin, purchasing, cashier

Step-by-Step Checkout

1. Select Products

```
┌──────────────────────────────────────────┐
│ 🔍 Search products...                    │
├──────────────────────────────────────────┤
│ [All] [Tires] [Batteries] [Oils] [...] │
├──────────────────────────────────────────┤
│ ┌──────────────────┬──────┬─────────┐   │
│ │ Product Name     │ Price │ Stock   │   │
│ ├──────────────────┼──────┼─────────┤   │
│ │ ACCELERA Tire    │ ₱2058│   88    │   │
│ │ MOTOLITE Battery│ ₱4402│   45    │   │
│ │ PETRONAS Oil     │ ₱455 │  120    │   │
│ └──────────────────┴──────┴─────────┘   │
└──────────────────────────────────────────┘
```

Actions: - Click a product to add to cart - Use category filters to narrow down - Search by product name or code

2. Review Cart

```
┌──────────────────────────────────────────┐
│ 🛒 CART                      Total: ₱0 │
├──────────────────────────────────────────┤
│ [Product]      [Qty]    [Price]  [Remove] │
│ ACCELERA Tire    2     ₱4,116     [X]     │
├──────────────────────────────────────────┤
│ Discount: [____] %                        │
│                                           │
│            [CHECKOUT]                     │
└──────────────────────────────────────────┘
```

3. Checkout Process

Click **CHECKOUT** to open the checkout modal:

```
┌──────────────────────────────────────────┐
│ CHECKOUT                                 │
├──────────────────────────────────────────┤
│ Customer: [Select or enter guest...]     │
│                                          │
│ Service Advisor: [Name field]            │
│                                          │
│ Receipt Type: (•) SI   ( ) DR            │
│                                          │
│ TIN: [____________]                      │
│ Business Address: [_________________]    │
│                                          │
│ Withholding Tax: [__] %                  │
│                                          │
│ Payment: (•) Cash   ( ) Card   ( ) GCash │
├──────────────────────────────────────────┤
│ Subtotal:     ₱4,116                     │
│ Discount:     ₱0                         │
│ Tax:          ₱0                         │
│ ________________________                 │
│ TOTAL:        ₱4,116                     │
├──────────────────────────────────────────┤
│        [PRINT RECEIPT]  [COMPLETE]       │
└──────────────────────────────────────────┘
```

Fields: - **Customer**: Select from existing or enter as guest - **Service Advisor**: Optional field for service jobs - **Receipt Type**: SI (Sales Invoice) or DR (Delivery Receipt) - **TIN/Business Address**: For business invoices - **Payment Method**: Cash, Card, or GCash

4. Complete Transaction

1. Click **COMPLETE** to process the sale
2. Receipt will be printed automatically
3. Stock will be deducted automatically
4. Order is recorded in the system

Printing Receipts

- **SI Receipt**: Standard sales invoice
- **DR Receipt**: Delivery receipt for item transfers
- Print options available after completion

Products Management

Manage your product catalog with full details including pricing, variants, and specifications.

Access

- **URL**: `/products`
- **Roles**: super_admin, admin, purchasing, purchaser

Product Fields

Field	Description	Required
Name	Product name	✅
Description	Product details	❌
Category	Product category	✅
Brand	Product brand	✅
Price	Selling price	✅
Cost Price	Purchase cost	❌
Size	Size/variant (e.g., 225/45 R17)	❌
Stock	Initial stock level	❌
Is Service	Mark as service item	❌

Tire-Specific Fields

Field	Description
PCD	Pitch Circle Diameter
Offset ET	Wheel offset
Width	Section width
Bore	Center bore
Finish	Wheel finish
Speed Rating	Speed rating (e.g., V, W, Y)
Load Index	Load capacity index
DOT Code	Department of Transportation code
Ply Rating	Ply rating

Managing Products

Add New Product

1. Click **+ NEW PRODUCT**
2. Fill in the product details form
3. Select category and brand
4. For tires, fill in tire-specific fields
5. Click **SAVE**

Edit Product

1. Find the product in the list
2. Click the **Edit** (pencil) icon
3. Modify the details
4. Click **UPDATE**

Delete Product

1. Find the product in the list
2. Click the **Delete** (trash) icon
3. Confirm the deletion
4. Product is soft-deleted (can be recovered)

Product Variants

Products can have variants linked via **Parent ID**:

1. Create a parent product (e.g., "TIRE 225/45 R17")
2. Create variants with size differences
3. Link variants to parent product

Inventory Management

Complete inventory control with batch tracking, stock movements, and warehouse management.

Access

- **URL**: `/inventory`
- **Roles**: super_admin, admin, purchasing, purchaser

Tabs

1. Stock Levels

View current stock across all warehouses.

```
┌──────────────────────────────────────────────────┐
│ Stock Levels                                     │
├──────────────────────────────────────────────────┤
│ Warehouse: [All Warehouses ▼] [Export]           │
├──────────────────────────────────────────────────┤
│ Product         | Warehouse | Stock | In Transit |
├─────────────────┼───────────┼───────┼────────────┤
│ ACCELERA Tire   | Main      |  88   |     0      |
│ MOTOLITE Batt   | Main      |  45   |    10      |
└─────────────────┴───────────┴───────┴────────────┘
```

Features: - Filter by warehouse - View in-transit stock - Export to Excel

2. Receive Stock (IN)

Record incoming inventory from suppliers.

Step-by-Step:

1. Click **+ NEW RECEIVE**
2. Select **Supplier**

3. Select **Warehouse** (your branch warehouse)
4. Add products with:
5. Product name
6. Quantity received
7. Unit cost
8. Enter **Reference** (e.g., PO number, delivery receipt)
9. Click **RECEIVE**

```
┌──────────────────────────────────────────┐
│ Receive Stock                            │
├──────────────────────────────────────────┤
│ Supplier: [Select Supplier ▼]            │
│ Warehouse: [Main Warehouse ▼]            │
│ Reference: [PO-2024-001]                |
├──────────────────────────────────────────┤
│ Products:                                │
│ ┌──────────────────────────────────────┐ │
│ │ [ACCELERA Tire    ] Qty: [10] Cost: | |
│ │ [MOTOLITE Battery ] Qty: [5]  Cost:  | |
│ └──────────────────────────────────────┘ |
│ [+ Add Item]                             |
├──────────────────────────────────────────┤
|          [CANCEL]  [RECEIVE STOCK]       |
└──────────────────────────────────────────┘
```

3. Stock Out (OUT)

Record stock removals (damaged, expired, adjustments).

Step-by-Step:

1. Click **+ NEW STOCK OUT**
2. Select **Product**
3. Enter **Quantity**
4. Enter **Reference** (reason)
5. Click **SUBMIT**

4. Movement Logs

View complete history of all stock movements.

Includes: - Date and time - Product name - Movement type (IN/OUT/ADJUSTMENT) - Quantity changed - Reference number - User who made the change

Batch Tracking

Each stock movement is tracked in batches:

- **Batch Number**: Auto-generated or manual
- **Expiry Date**: For products with expiration
- **Unit Cost**: Cost per unit at time of receipt
- **追溯**: Full traceability of stock

Branches & Transfers

SMSystem supports multi-branch operations with isolated inventory.

Access

- **URL**: `/branches` (admin only)
- **URL**: `/transfers` (all users)

Branch Management (Admin)

Add New Branch

1. Go to **Branches**
2. Click **+ NEW BRANCH**
3. Enter branch details:
4. Branch Name
5. Branch Code (unique identifier)
6. Address
7. Phone Number
8. Click **SAVE**

Transfer System

Transfers allow moving stock between branches.

Transfer Flow

```
[Source Branch] ──request──> [Request Sent]
                                    │
                                    ▼
                            [Destination Branch]
                                    │
                                    ▼ (approve/reject)
                              [Approved/Rejected]
                                    │
                                    ▼ (if approved)
                               [In Transit]
                                    │
                                    ▼
                             [Received at Destination]
```

Creating a Transfer Request

Step-by-Step:

1. Go to **Transfers**
2. Click **+ REQUEST STOCK**
3. Select **Source Branch** (the branch to request from)
4. Add products and quantities
5. Add notes (optional)
6. Click **SUBMIT REQUEST**

```
┌──────────────────────────────────────────┐
│ Request Stock from Another Branch         │
├──────────────────────────────────────────┤
│ Request From: [Select Branch ▼]          │
│                                          │
│ Products:                                │
│ ┌──────────────────────────────────────┐ │
│ │ [Product Name]        Qty: [___] X   │ │
│ └──────────────────────────────────────┘ │
│ [+ Add Product]                          │
│                                          │
│ Notes: [Optional notes for request...]   │
├──────────────────────────────────────────┤
│        [CANCEL]  [SUBMIT REQUEST]        │
└──────────────────────────────────────────┘
```

Managing Transfer Requests

As Requester (Destination Branch):

Status	Meaning	Action
pending	Awaiting source branch approval	Cancel request
rejected	Source branch rejected	View reason
approved	Source approved, awaiting shipment	Wait for shipping
in_transit	Shipped, in delivery	Wait for receipt
completed	Received at destination	Done

As Source Branch:

1. View incoming requests in **Incoming** tab
2. Review requested items
3. **Approve**: Accept the transfer
4. **Reject**: Decline with reason
5. **Ship**: Mark as in_transit (this deducts from your stock)
6. Once shipped, await destination receipt confirmation

Receiving Transfers

When stock arrives at destination:

1. Go to **Transfers** → **Incoming**
2. Find transfer with **in_transit** status
3. Click **RECEIVE**
4. Confirm receipt
5. Stock is added to destination warehouse

Orders Management

View and manage all sales orders.

Access

- **URL**: `/orders`
- **Roles**: All users

Order Statuses

Status	Description
pending	Order created, awaiting payment
completed	Payment received, fulfilled
cancelled	Order cancelled
refunded	Payment refunded

Viewing Orders

Order List: - Filter by status - Search by order ID or customer - View order details - Print receipts (SI or DR) - Process refunds

Order Details: - Customer information - List of items purchased - Payment method - Total amounts - Timestamp

Printing Receipts

1. Find the order
2. Click **View** (eye icon)
3. Click **PRINT RECEIPT** or **PRINT DR**

Customers & CRM

Manage customer relationships and track customer purchases.

Access

- **URL**: `/customers`
- **URL**: `/crm` (admin only)
- **Roles**: All users for basic; Admin for CRM

Customer Management

Add Customer

1. Go to **Customers**
2. Click **+ NEW CUSTOMER**
3. Fill in details:

4. Name
5. Phone Number
6. Email (optional)
7. Address (optional)
8. TIN (for business customers)
9. Click **SAVE**

Customer Fields

Field	Description
Name	Customer's full name or business name
Phone	Primary contact number
Email	Email address
Address	Billing/shipping address
TIN	Tax Identification Number
Total Purchases	Accumulated purchase value
Order Count	Number of orders placed
Last Visit	Date of last purchase

CRM Features (Admin)

URL: `/crm`

Customer Analytics

- **Top Customers**: By total purchases
- **Recent Activity**: Latest customer interactions
- **Loyalty Tracking**: Monitor repeat customers

Expenses

Track and categorize business expenses.

Access

- **URL**: `/expenses`

- **Roles**: super_admin, admin, purchasing, purchaser

Adding Expenses

1. Click **+ NEW EXPENSE**
2. Fill in details:
3. Description
4. Amount
5. Category
6. Date
7. Notes (optional)
8. Attach receipt (optional)
9. Click **SAVE**

Expense Categories

Default categories: - Utilities - Supplies - Maintenance - Transportation - Miscellaneous

Reports

View expenses by: - Date range - Category - Total amounts

Suppliers & Purchase Orders

Manage suppliers and create purchase orders.

Access

- **URL**: `/suppliers`
- **URL**: `/purchase-orders`
- **Roles**: super_admin, admin, purchasing, purchaser

Supplier Management

Add Supplier

1. Go to **Suppliers**
2. Click **+ NEW SUPPLIER**
3. Enter details:
4. Company Name

5. Contact Person
6. Phone
7. Email
8. Address
9. Click **SAVE**

Purchase Orders

Create purchase orders to request stock from suppliers.

Creating a PO

1. Go to **Purchase Orders**
2. Click **+ NEW PO**
3. Select **Supplier**
4. Add products with quantities and expected costs
5. Set **Expected Delivery Date**
6. Click **SUBMIT**

PO Statuses:

Status	Description
pending	Awaiting supplier confirmation
approved	Supplier approved
shipped	Supplier shipped
received	Stock received at warehouse
cancelled	PO cancelled

Receiving Against a PO

1. Open the approved/shipped PO
2. Click **RECEIVE STOCK**
3. Verify quantities match
4. Confirm receipt
5. Stock is added to warehouse

Staff Management

Manage user accounts and permissions.

Access

- **URL**: `/staff`
- **Roles**: super_admin, admin only

Adding Staff

1. Go to **Staff**
2. Click **+ NEW STAFF**
3. Fill in details:
4. Name
5. Email
6. Password
7. Role
8. Branch assignment
9. Click **SAVE**

User Fields

Field	Description
Name	Staff's full name
Email	Login email (unique)
Password	Initial password
Role	User permission level
Branch	Assigned branch
Status	Active/Inactive

Role Assignment

Choose appropriate roles based on job function: - **Admin**: Branch manager - **Purchasing**: Inventory manager - **Cashier**: Sales associate - **User**: Basic access

Reports & Logs

Daily Reports

URL: `/daily-report`
Roles: admin only

View comprehensive daily reports including: - Total sales amount - Number of transactions - Top-selling products - Payment method breakdown - Low stock alerts

Activity Logs

URL: `/logs`
Roles: admin only

Track all system activities: - User logins - Product changes - Inventory movements - Order processing - Settings changes

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Settings

Configure system-wide settings.

Access

- **URL**: `/settings`
- **Roles**: admin only

Available Settings

Setting	Description
Business Name	Your company name
Business Address	Default address on receipts
TIN	Your Tax Identification Number
Service Advisors	List of service advisor names
Printer Name	Receipt printer device name
Invoice Prefix	Prefix for invoice numbers
Receipt Footer	Custom footer message

API Documentation

SMSystem provides a RESTful API for integrations.

Base URL

```
http://localhost:8080/api
```

Authentication

All API requests require a JWT token:

```
Authorization: Bearer <your-token>
```

Endpoints

Authentication

Method	Endpoint	Description
POST	`/auth/login`	User login
POST	`/auth/register`	User registration
GET	`/auth/me`	Get current user

Products

Method	Endpoint	Description
GET	`/products`	List products
POST	`/products`	Create product
GET	`/products/:id`	Get product
PUT	`/products/:id`	Update product
DELETE	`/products/:id`	Delete product

Orders

Method	Endpoint	Description
GET	`/orders`	List orders
POST	`/orders`	Create order
GET	`/orders/:id`	Get order
PUT	`/orders/:id`	Update order

Inventory

Method	Endpoint	Description
GET	`/inventory/levels`	Stock levels
POST	`/inventory/in`	Receive stock
POST	`/inventory/out`	Stock out
GET	`/inventory/logs`	Movement logs

Transfers

Method	Endpoint	Description
GET	`/transfers`	List transfers
POST	`/transfers`	Create transfer
PUT	`/transfers/:id`	Update transfer

Example Request

```
# Login
curl -X POST http://localhost:8080/api/auth/login \
  -H "Content-Type: application/json" \
  -d '{"email":"admin@example.com","password":"password"}'

# Response
{
  "token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...",
  "user": { "id": 1, "name": "Admin", "email": "admin@example.com" }
}

# Use token for authenticated requests
curl http://localhost:8080/api/products \
  -H "Authorization: Bearer eyJhbGciOiJIUzI1NiIs..."
```

Troubleshooting

Common Issues

Login Problems

Issue: Cannot log in **Solutions**: 1. Verify email and password are correct 2. Check if account is active 3. Clear browser cache and try again 4. Contact admin to reset password

Stock Discrepancies

Issue: Stock levels don't match **Solutions**: 1. Check pending transfers (in_transit status) 2. Review movement logs for unauthorized changes 3. Verify all receipts were processed 4. Check for data sync issues

Printer Not Working

Issue: Receipt won't print **Solutions**: 1. Check printer is connected and powered on 2. Verify printer name in Settings 3. Check browser's print permissions 4. Try printing to PDF as workaround

Slow Performance

Issue: System is slow **Solutions**: 1. Check internet connection 2. Clear browser cache 3. Close unused browser tabs 4. Refresh the page

Error Messages

Error	Meaning	Solution
"Unauthorized"	Invalid or expired token	Log in again
"Forbidden"	Insufficient permissions	Contact admin
"Not found"	Resource doesn't exist	Check URL
"Server error"	Backend issue	Contact support

Data Backup

Regular backups are recommended: 1. MySQL database: Use mysqldump 2. Settings: Export from Settings page 3. Store backups in a secure location

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Support

For issues or questions: - **Email**: support@smsystem.com - **GitHub**: https://github.com/Jayem09/SMSystem - **Documentation**: https://docs.smsystem.com

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Changelog

v0.3.0 (March 2026)

- Fixed stock-per-branch isolation
- Security patches for SQL injection
- UI improvements (DataTable, StatusBadge)
- UX enhancements

v0.2.x (Previous versions)

- Initial releases with core features
- POS system
- Inventory management
- Multi-branch support

SMSystem v0.3.0 - Documentation generated March 2026